International Journal of Service and Knowledge Management
International Institute of Applied Informatics
2026, Vol. 10, No. 1, IJSKM940

Time-Efficient Multi-Agent Simulation Reflecting Actual Resident Population

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Abstract

This study proposes a Time-Efficient Multi-Agent Simulation (Time-Efficient MAS) for evaluating the social impacts of transportation policies. The model generates agents reflecting actual population composition based on person-trip survey data and reconstructs realistic travel routes using GIS data, reproducing mobility patterns. A distinctive feature of this study is the introduction of two novel analytical units: Co-Exist, representing spatiotemporal coexistence, and Repeated Chance Meetings (RCM), indicating coincidental opportunities for contact among the same individuals. Through these concepts, the study enables quantitative analysis of opportunity structures for contact, which has been difficult to assess in conventional transportation research. By separating route computation between target and surrounding areas and using a regression-based performance model that predicts computation time from network size prior to execution, the framework significantly reduces computational load while maintaining spatial accuracy, enabling month-scale simulations of residents' behavior. In a case study focusing on the Tomioka district of Yokohama City, results revealed that bus route reductions and population aging tend to localize residents' activity ranges and decrease opportunities for social contact. This study highlights transportation as social infrastructure shaping opportunities for human contact, providing knowledge to support transportation policy design and evidence-based decision-making.

Keywords: multi-agent simulation, person-trip survey, time–space cube, co-move and co-stay, repeated chance meetings

1 Introduction

1.1 Motivation and Problem Statement

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In regional communities facing population decline and aging, the reduction of public transportation services has affected both daily activities and social interactions. The discontinuation or reduction of bus routes has led to fewer opportunities for going out, gradually diminishing residents' interactions and spontaneous opportunities for contact. Transportation, therefore, should not be viewed merely as a means of movement but as a form of social infrastructure that connects people, with its decline potentially influencing Social Capital (SC) [1] and Subjective Well-being (SWB) [2]. However, conventional transportation studies have primarily focused on physical indicators such as efficiency and profitability, leaving the social value of transportation largely unexamined. To address this issue, the present study focuses on Co-Exist, representing the shared spatiotemporal presence of individuals, and Repeated Chance Meetings (RCM), denoting recurring accidental opportunities for contact, to quantitatively reassess the social value generated by public transportation. Conventional surveys and aggregated mobility statistics can capture population-level movement patterns, but they have difficulty representing the recurrent temporal structure of contact opportunities among the same individuals. To express such history-dependent contact structures, this study adopts a Multi-Agent Simulation (MAS) framework.

1.2 Research Objectives and Scope

The objective of this study is to quantitatively examine how transportation policies influence the structure of social interactions among local residents.
(1) To define Co-Exist and RCM as new social indicators and to establish a method for their quantitative estimation using a MAS [3] framework.
(2) To develop a Time-Efficient MAS that enables the execution of large-scale urban simulations within a realistic computation time.
The proposed Time-Efficient MAS separates route computation between target and surrounding areas, reducing computational cost while maintaining accuracy and enabling efficient comparison of transportation policies. To achieve these objectives, the simulation must handle approximately 18,000 agents over a 30-day period, which requires a substantial reduction in route computation time to make the model practically executable.

1.3 Contributions

This study provides two main contributions to the modeling and analysis of community-level mobility and social interaction.
(1) The first contribution is the introduction of Co-Exist and RCM as quantitative indicators for capturing the spatial and temporal structure of residents' co-presence and repeated encounters. By extending a MAS framework to explicitly compute these indicators, the study establishes a methodological basis for assessing how transportation systems influence social-contact opportunities in small-scale urban areas.
(2) The second contribution is the development of a Time-Efficient MAS capable of executing large-scale simulations within practical computation time. The proposed method reduces route computation cost by separating searches for the target and surrounding areas and by applying a linear computational model derived from empirical measurement. This framework enables month-scale simulations involving approximately 18,000 agents, which would otherwise be computationally infeasible.
Together, these contributions present a unified approach for quantitatively evaluating both the social-interaction structure and the computational feasibility of mobility simulations at the community scale.

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1.4 Paper Organization

This paper is structured around two main themes: the quantification of social contact opportunities in regional transportation and the acceleration of MAS.
Chapter 2 reviews prior studies on multi-agent simulation, human mobility analysis, and related work on Co-Exist and RCM, clarifying the theoretical background and research gap of this study. Chapter 3 defines the concepts of Co-Exist and RCM and presents their theoretical framework. Chapter 4 presents the overall architecture of the Time-Efficient MAS and its computational acceleration method. Chapter 5 describes the experimental design and case study settings for the Keikyu Tomioka district. Chapter 6 presents the results and discusses how transportation policies influence social interaction structures using Co-Exist and RCM. Chapter 7 concludes the study and outlines its limitations and future research directions.

2 Related Work and Background

2.1 Mobility Simulation for Urban Systems

2.1.1 Engineering Approach: Agent-Based Urban Mobility Models

MAS has evolved as a representative approach for modeling urban transportation and social behavior, enabling the analysis of complex system dynamics. In MAS, residents are modeled as autonomous agents, and their interactions reproduce traffic flows and social behaviors. MATSim [4] and VISSIM™ are representative platforms of this approach, widely used for analyzing traffic demand, commuting behavior, and transportation policies. However, many existing studies have focused on physical indicators such as traffic efficiency and waiting time, without sufficiently addressing human social behavior, psychological factors, and interpersonal relationships within local communities.

2.1.2 Social and Psychological Perspectives on Mobility and Interaction

Recent research has advanced the perspective that transportation should be understood not merely as a means of movement, but as a social process that connects people. Oldenburg (1999) [5] highlighted the importance of "third places", where spontaneous opportunities for contact foster community formation, emphasizing that urban public spaces serve as venues for generating social connections. Putnam (2000) [6] argued that interactions through shared travel experiences and local events enhance SC and SWB, and Gehl (1987) [7] demonstrated that urban design can stimulate people's staying behaviors and social interactions. In the field of psychology, Zajonc (1968) [8] proposed the mere exposure effect, demonstrating that repeated opportunities for contact foster familiarity and trust, and Hall (1966) [9] introduced the concept of proxemics, which defines how physical distance between individuals shapes social relationships. These studies suggest that mobility forms the foundation of human psychological and social interactions; however, most remain qualitative.

2.2 Human Flow and Data Platforms

2.2.1 Utilization and Significance of Human Mobility Data

To apply MAS to real-world societies, it is essential to construct behavioral models based on actual human mobility data. In recent years, studies have advanced that acquire and anonymize

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individuals' movement trajectories and dwelling times in chronological order to analyze regional mobility patterns. Representative examples include the PFLOW Project [10] by the Center for Spatial Information Science at the University of Tokyo and the Human Flow Project [11] by NTT DOCOMO's Mobile Spatial Statistics.

2.2.2 Characteristics of Existing Data Infrastructures

The PFLOW Project generates synthetic population flows based on census and transportation survey data for academic analysis, while Mobile Spatial Statistics visualize regional population distributions and mobility trends using anonymized mobile phone data.

Table 1: Projects on Mobility Analysis Focusing on Human Movement

Category	PFLOW Project	Human Flow Project
Objective	Academic research and addressing social issues	Human mobility analysis for commercial and administrative purposes
Activity	Academic (research outcomes are publicly available)	Practical (results provided to clients or government agencies)
Fields	Urban planning, disaster prevention, tourism promotion	Commercial analysis, administrative decision support, disaster response, infectious disease control

2.2.3 Limitations and Challenges

While these data infrastructures capture citywide population dynamics, they are limited to macro-level analysis and cannot adequately reproduce micro-level social behaviors such as individual interactions and decision-making. In addition, reproducibility declines in sparsely populated areas or contexts influenced by cultural and psychological factors.

2.3 Research Gap and Direction of This Study

Existing studies emphasize social interactions in transportation and urban design, but lack a framework to quantitatively reproduce opportunities for contact over time. Moreover, city-scale MAS faces excessive computational costs due to the growth of agents and route searches, limiting policy comparison within practical computation time. To address these issues, this study proposes a Time-Efficient Multi-Agent Simulation that reflects actual population composition to quantitatively evaluate the impact of transportation policies on social contact opportunities.

3 Conceptual Framework of Co-Exist and Repeated Chance Meetings (RCM)

Mobility in urban environments is a social process in which people share time and space. This chapter introduces the concepts of Co-Exist, which integrates Co-Move and Co-Stay [12], and RCM, and presents a conceptual framework for redefining mobility.

3.1 Conceptual Overview

Conventional urban mobility studies have focused on transportation efficiency and route optimization, while social interactions emerge through the sharing of time and space. Although such

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contact opportunities influence SC and SWB, their spatiotemporal structures have not been quantitatively captured. This section introduces the concepts of Co-Exist and RCM as a basis for redefining mobility as a structure that generates social contact and for the MAS presented in the next chapter.

3.2 Co-Exist

Co-Exist refers to the condition in which multiple individuals share the same physical space during a specific time period. It represents the spatiotemporal overlap of human presence itself and enables the quantitative identification of where and when groups of people emerge within urban spaces. In this study, two spatial phenomena that constitute Co-Exist are defined as follows:

- Co-Move: Refers to the parallel behavior of individuals moving in the same direction, typically observed in contexts such as commuting trains, buses, or pedestrian flows.
- Co-Stay: Refers to the condition in which individuals share the same location or a confined area for a certain duration, representing static co-presence in places such as station buildings, commercial facilities, or parks.

The distribution and duration of these spatial overlaps form a precondition for RCM. In this study, Co-Exist is defined as a pair of agents simultaneously occupying positions within a specified spatiotemporal distance, and its statistical properties are used to analyze how urban structures and transportation networks shape opportunities for social contact.

3.3 Repeated Chance Meetings (RCM)

Repeated Chance Meetings (RCM) refer to the recurrence of Co-Exist occurrences between the same individuals over time, representing the continuity of repeated opportunities for contact rather than a single instance of coexistence. RCM quantitatively captures re-opportunities for contact in urban environments and the latent processes of social relationship formation. This phenomenon corresponds to the mere exposure effect proposed by Zajonc (1968), which suggests that repeated opportunities for contact foster familiarity and trust even without intentional interaction. Therefore, RCM is positioned not as an occurrence of contact opportunity, but as its recurrence and continuity. In this study, RCM is defined as the case in which a pair of agents that once coexisted within a given spatiotemporal domain coexist again at another time. This definition enables quantitative evaluation of the frequency, periodicity, and duration of repeated co-occurrence, providing insights into how transportation policies and urban structures shape patterns of repeated opportunities for contact.

3.4 Social Implications of Co-Exist and Repeated Chance Meetings (RCM)

Co-Exist and RCM capture the structural patterns of spontaneous contact opportunities and their recurrence in urban environments. Their significance lies not merely in shared space, but in the potential for repeated encounters to develop into social relationships. Repeated opportunities for contact contribute to the accumulation of familiarity and trust, which supports the formation of SC and is associated with SWB. Accordingly, mobility can be redefined not only as the optimization of movement efficiency, but as a relational infrastructure that supports social well-being. The MAS proposed in this study provides a quantitative framework for analyzing how such structures of contact and re-contact emerge in urban environments.

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4 Proposed Method: Time-Efficient MAS Framework

This chapter outlines the design concept of a Time-Efficient Multi-Agent Simulation (Time-Efficient MAS) developed to quantitatively reproduce Co-Exist and RCM in urban mobility contexts. To achieve feasible long-term simulation at the city scale, the framework adopts a separated route-computation structure for target and surrounding areas, balancing computational efficiency with modeling accuracy.

4.1 Concept of Time-Efficient MAS

The Time-Efficient MAS developed in this study reproduces long-term movements of an entire local population and quantitatively measures Co-Exist and RCM under limited computational resources. Its core design adopts a spatially hierarchical structure, applying detailed route computation in the target area and simplified movement in the surrounding area to balance accuracy and efficiency (Figure 1).

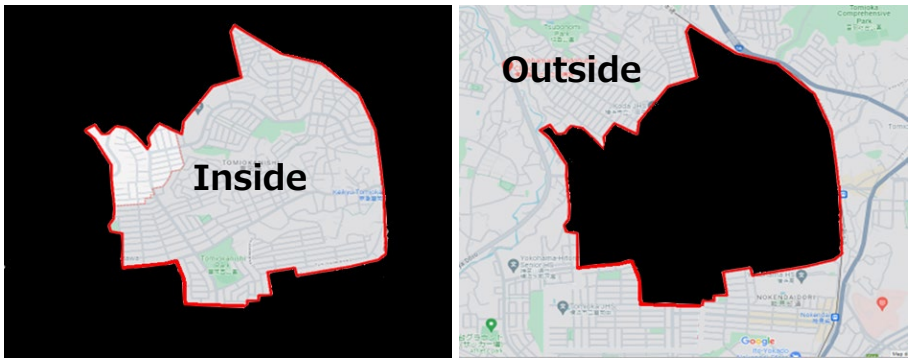


Figure 1: Spatial Division of Target and Surrounding Areas

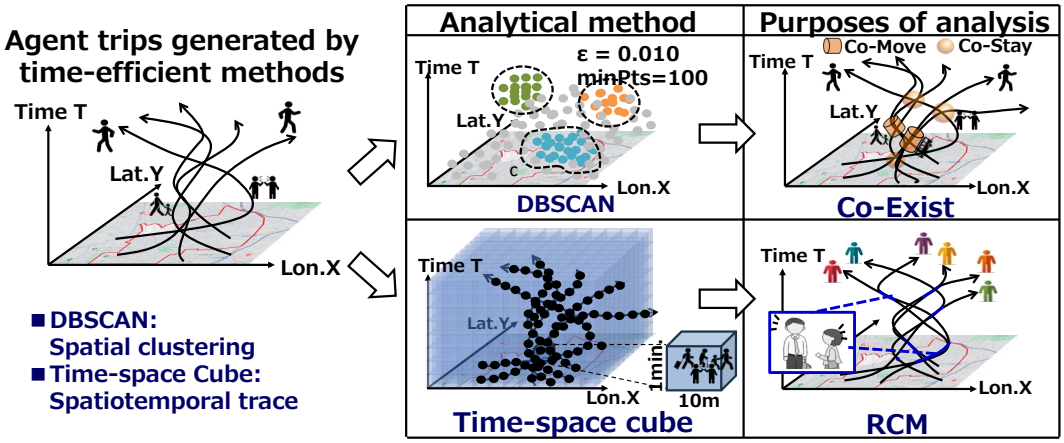


Figure 2: Analytical Process for Co-Exist and RCM Detection

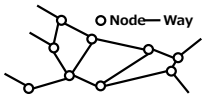
Within the target area, agents' movement histories are recorded with time, coordinates, and transport mode. Using the DBSCAN clustering algorithm [13] agents simultaneously occupying the same spatiotemporal range are extracted as Co-Exist events. A Time-Space Cube structure is then used to detect and quantify cases in which the same agent pairs form Co-Exist events at different times and locations as RCM events (Figure 2).

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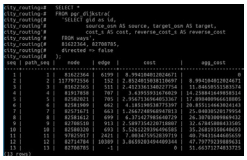
4.2 Route Calculation Separation Between Target and Surrounding Areas

Within the target area, agents can use all local transportation modes, including buses and on-demand services. Public transport follows fixed timetables, while walking, bicycle, and motorbike movements follow shortest-path routes computed using the Dijkstra algorithm [14] implemented in the PostGIS [15] extension of PostgreSQL [16] (pgr_dijkstra). Although Dijkstra's theoretical complexity is $O((V + E) \log V)$, where V denotes the number of nodes and E the number of edges [17]. However, in a PostGIS environment, implementation-dependent overheads such as SQL processing and I/O latency increase computation time beyond theoretical estimates. To account for these effects, this study introduces a hypothetical execution time model (Figure 3).

$$T(V,E) = \alpha\,(V+E)\,\log V + \beta \qquad (1)$$



V: Node
E: Edge (Way)

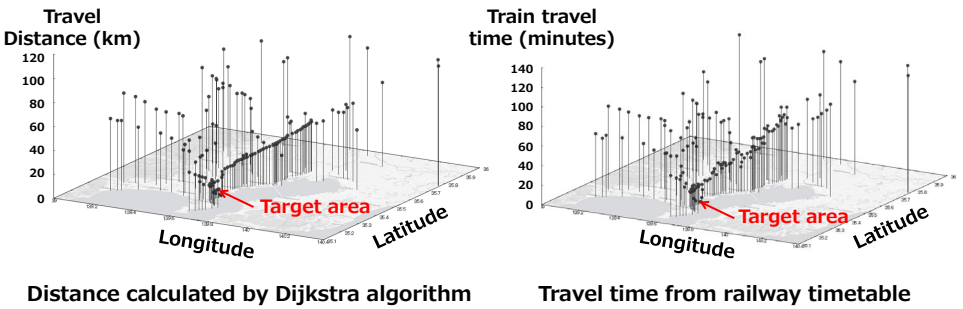




If PostGIS use..	Complexity	Computation Time Mode
No Priority Queue	$O(V^2)$	$\boldsymbol{T(V) = \alpha' V^2 + \beta'}$
With Priority Queue	$O\,((V+E)\,\log V)$	$\boldsymbol{T(V,E) = \alpha\,(V+E)\,\log V + \beta}$

Figure 3: Hypothetical Execution Time Model for Route Computation in PostGIS

Here, α represents the algorithmic computation coefficient of the Dijkstra method, while β denotes a constant reflecting PostGIS-specific system overhead. This hypothetical model combines theoretical complexity with implementation-dependent delays and is validated in Chapter 5 using actual route-search results. The surrounding area is defined as a boundary zone for reproducing inflow and outflow between the target area and its exterior. To reduce computational load, simplified rule-based routing is applied instead of detailed path computation, focusing on reproducing overall mobility flow. Precomputed shortest paths between major nodes are reused to avoid real-time recalculation. For efficient estimation of long-distance travel time and distance in the surrounding railway area, bilinear interpolation [18] based on data from 167 stations is employed (Figure 4).



Distance calculated by Dijkstra algorithm **Travel time from railway timetable**

Figure 4: Bilinear Interpolation Model in the Surrounding Area

This method estimates the value at an interior point based on four known data points on a spatial grid. By selecting one reference point from each of the four quadrants surrounding the destination,

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directional bias is avoided. This approach preserves the spatial continuity of estimated values while ensuring smooth variation across the interpolation domain (Figure 5).

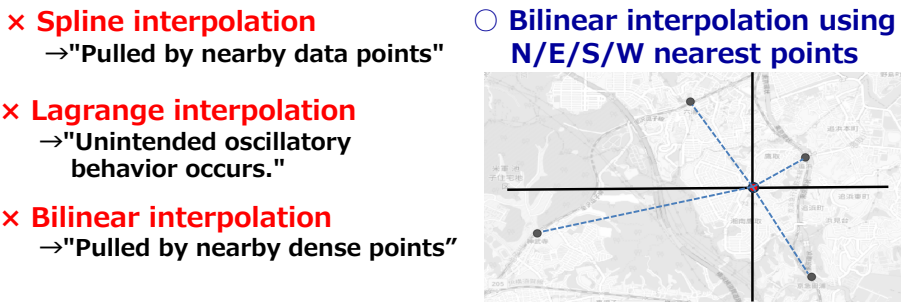


Figure 5: Concept of Bilinear Interpolation with Four Reference Points

4.3 Implementation and Processing Flow

4.3.1 Route Computation

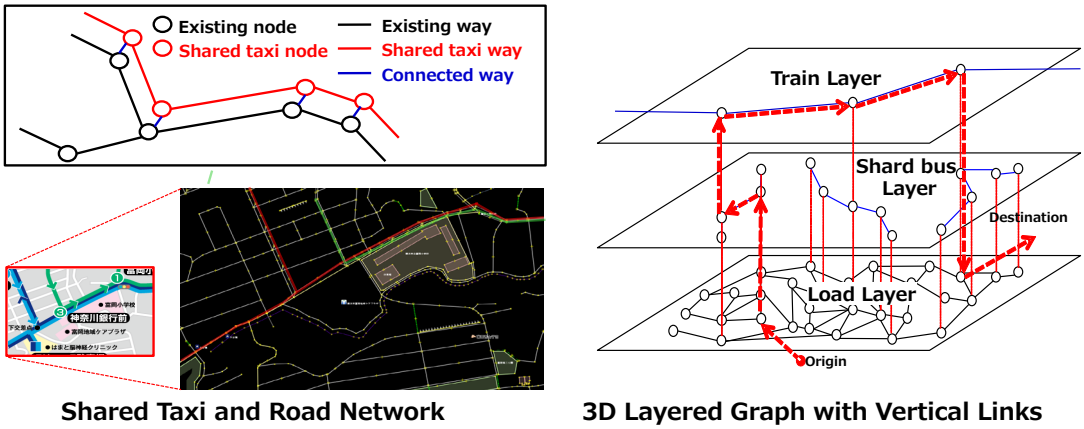


Figure 6: 3D Multi-layer Network Integrating Road and Shared Taxi Routes

As shown in Figure 6, the system adopts a multi-layered structure that connects shared-taxi routes with the road network through inter-layer links, forming a three-dimensional route framework. Routes for walking, bus, railway, and shared taxi are computed independently and integrated to enable efficient multi-modal routing. This unified structure allows time-efficient and accurate reproduction of mobility within and beyond the target area.

4.3.2 Input Data and Preprocessing

The input data consist of person-trip (PT) survey records for the target area and GIS data. The PT survey provides resident attributes such as location, demographics, trip purpose, transport mode, departure time, and stay duration, while the GIS data include spatial information on transportation networks and urban facilities.

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4.3.3 Agent Generation

Figure 7 illustrates the generation of resident agents from person-trip (PT) records, adjusted to match the actual population size of the target area.

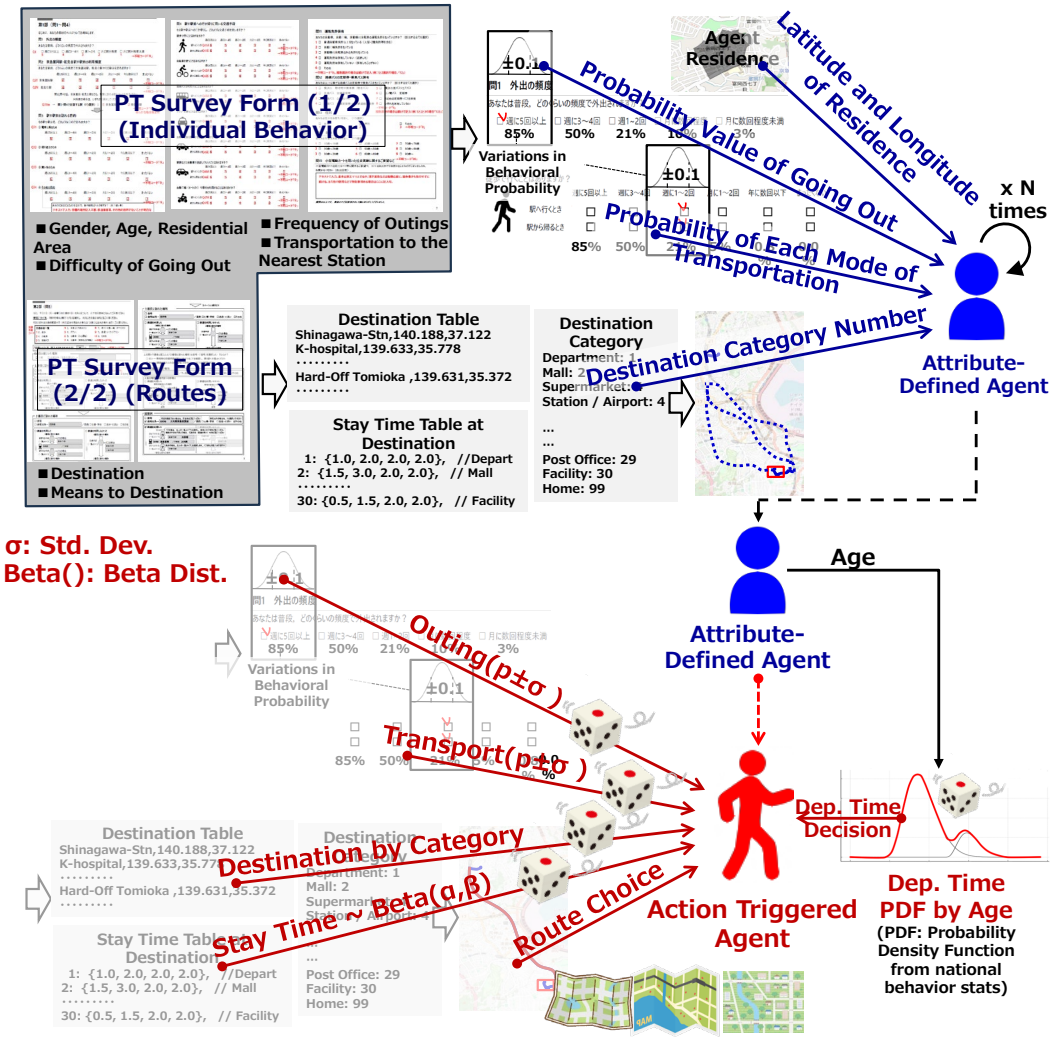


Figure 7: Generation Process of Attribute-Defined and Action-Triggered Agents

Each PT record is converted into a single agent retaining its attributes, including residential coordinates, trip purpose, departure time, transportation mode, and destination. At this stage, no probabilistic behavior is introduced, allowing later extension using probability distribution models described in Chapter 5.

5 Case Study: Tomioka Area, Yokohama City

In this chapter, the Time-Efficient MAS developed in the previous chapter is applied to the Tomioka district of Kanazawa Ward, Yokohama City, Kanagawa Prefecture. The validity of the proposed method is verified through simulations based on actual empirical data.

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5.1 Overview of Study Area and Data Sources

5.1.1 Target Area

The target area of this study is the Tomioka district located in Kanazawa Ward, Yokohama City, Kanagawa Prefecture. The district consists of nine subareas centered around Keikyu Tomioka Station, comprising a mix of residential zones, commercial areas, schools, and parks. Most of the area is hilly, featuring notable elevation differences, and local mobility is primarily conducted via short-distance modes such as walking, bicycling, and bus travel. The railway station is situated on the outer edge of the district, while internal movements within the area are mainly carried out by bus or on foot (Figure 8).

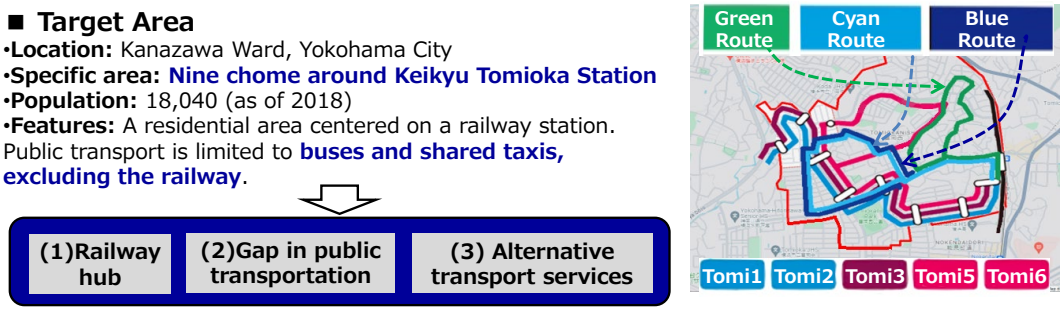


Figure 8: Target Area and Local Transport Network

The Tomioka district, characterized by limited public transportation accessibility, population decline, and progressive aging, represents a typical case of resident mobility in small urban areas. These characteristics make it an appropriate environment for empirical validation of social contact opportunities specifically, Co-Exist and RCM based on real-world data. The data used for agent generation are based on the results of a local person-trip (PT) survey conducted by Yokohama National University in the nine subareas surrounding Keikyu Tomioka Station following a field experiment (hereafter referred to as the Tomioka PT Survey) [19]. The survey targeted individuals aged 15 years and older, distributed to 6,628 households, and collected valid responses from 1,357 households, totaling 2,093 individuals. Each record is managed using a five-digit individual identification number, allowing consistent tracking of multiple trips made by the same person. The transportation mode codes are defined as follows: walking (1), bicycle (2), private car (3), motorcycle (4), taxi (5), route bus (6), shared taxi (7), railway (8), and others (9). For multi-modal trips, the corresponding codes are concatenated according to the survey instructions (e.g., bus + walk = "61"). Destination codes include categories such as home (1), work (2), school (3), shopping (4), pickup/drop-off (5), private affairs (6), and return home (9). Additional details or special notes could be entered in a free-text field provided in the questionnaire.

5.1.2 Applied Measurement Methods

(1) Co-Exist Measurement

To extract spatiotemporal overlaps (Co-Exist), the method identifies groups of agents located within a specified spatial range during the same time interval and treats these as instances of spatial coexistence. These groups are then clustered using DBSCAN (Density-Based Spatial Clustering of Applications with Noise) [13]. In this study, the two parameters required by DBSCAN neighborhood distance (ε) and minimum points (minPts) are consistently determined

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based on three criteria: theoretical requirements, observational resolution, and social interpretability (Figure 9).

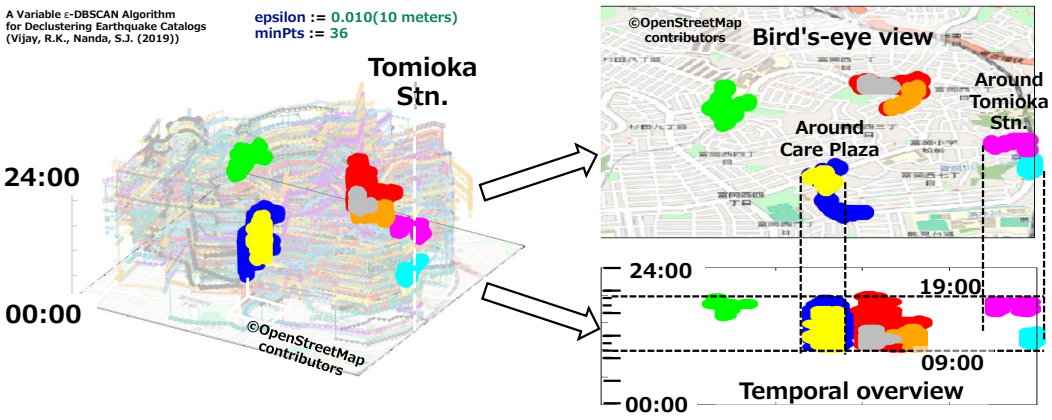


Figure 9: Criteria for Determining DBSCAN Parameters (ε and minPts)

After scaling the dataset so that the spatial axis corresponds to 1 meter units and the temporal axis to 30-second intervals, the parameters were set based on reference values derived from prior psychological experiments. Specifically, ε = 0.010 (corresponding to approximately 10 meters spatially and 14 minutes temporally) and minPts = 100, representing conditions in which 33 agents coexist for at least 1.5 minutes. These parameter settings reflect both the spatiotemporal resolution of observation and the socially meaningful scale of human interaction.

(2) RCM Measurement

The time space cube is a three-dimensional analytical framework composed of spatiotemporal cells, each representing a unit of 10 meters in spatial extent (east-west and north-south) and 1 minute in temporal duration. The 10-meter distance was adopted based on findings from psychological experiments identifying it as the threshold for maintaining visual contact [8], while the 1-minute interval was chosen as the minimum temporal unit during which individuals can recognize others even while in motion [20]. These values were therefore applied in this study as socially and behaviorally meaningful parameters. To quantitatively capture RCM generated through mobility, this study defines three original indicators (Figure 10).

(a) RCM Pair: A pair of agents that have shared the same spatiotemporal cell at least once. For example, if two agents coexist in the same location and time slot even once, they are recognized as an RCM pair.

(b) RCM Count: An indicator representing the total number of spatiotemporal cells in which contact occurred for each RCM pair. It reflects the frequency and spatialtemporal extent of repeated opportunities for contact. For instance, if a pair meets within three different cells, the RCM count is 3.

(c) RCM Day: An indicator expressing the number of days on which contacts occurred for a given RCM pair, allowing assessment of how many separate days opportunities for contact were sustained. For example, if contacts were recorded on June 7 and June 14, the RCM Day value is 2.

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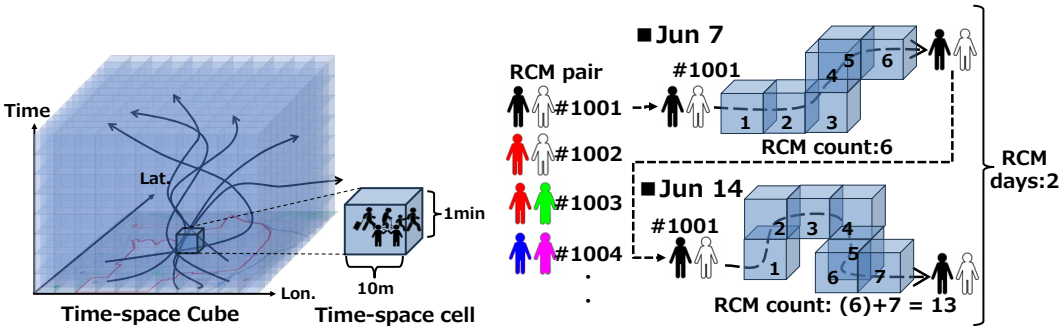


Figure 10: Time-Space Cube Framework for RCM Measurement

5.1.3 Targeted Transportation Policy Scenarios

Eight transportation policy scenarios were developed by combining two types of population scaling 2018 (observed) and 2045 (projected) with four different transportation policy conditions. Through these scenarios, the study quantitatively compares and evaluates how population scaling and transportation policies influence the occurrence and characteristics of RCM (Figure 11).

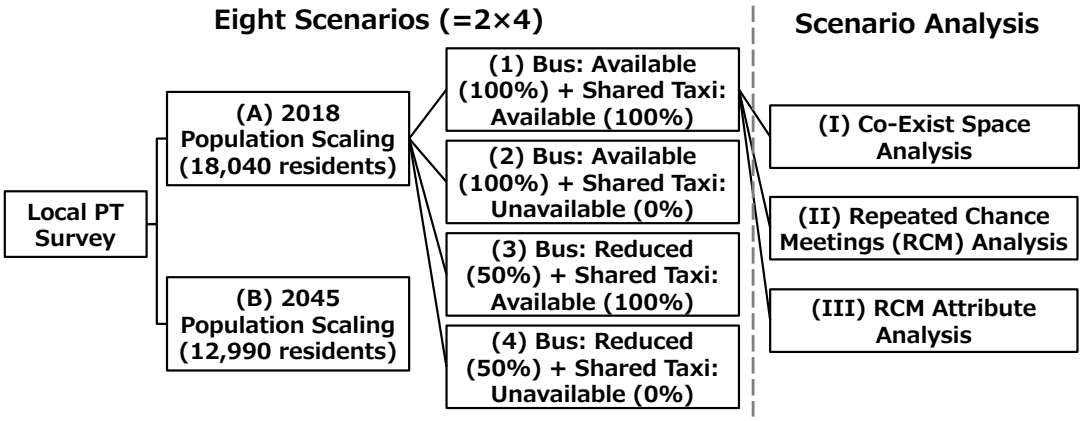


Figure 11: Analytical Scenarios Combining Population Scaling and Transportation Policies

5.2 Results and Discussion

5.2.1 Computational Efficiency of the Proposed Method

To evaluate the computational characteristics of the Dijkstra-based path search in PostGIS, this subsection examines how execution time scales with network size. The analysis follows the computational model introduced in Section 4.2 (Equation (1)). The model parameters were estimated using three Tomioka-area datasets: (i) tomioka_db_d, which adds regular bus routes; (ii) tomioka_db_e, which incorporates cart-path data; and (iii) tomioka_db_f, a simplified network retaining minimal connectivity. Figure 12 (left-side) plots these three datasets along with the fitted regression line. The regression yielded $\alpha=1.063\times10^{-7}$ and $\beta=7.465\times10^{-4}$, indicating a strong linear relationship that characterizes the dominant behavior of PostGIS Dijkstra computation in small networks.

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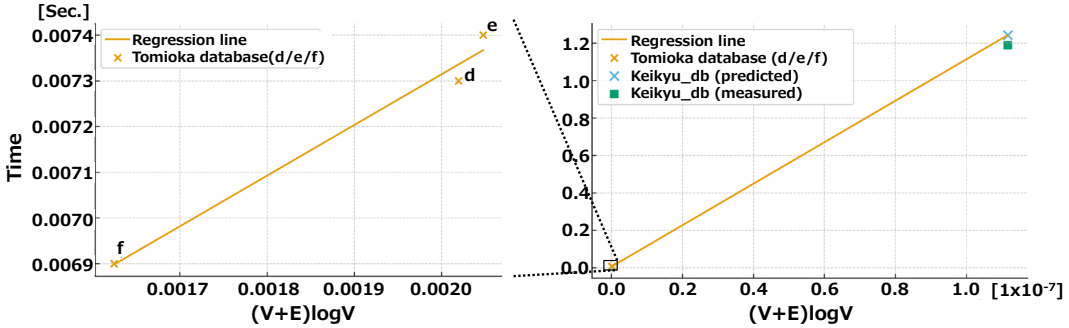


Figure 12: Regression on Small-Scale Networks and Extrapolation to a Large-Scale Network

Next, the model was applied to an independent large-scale network, keikyu_db (366,798 nodes; 504,186 edges), to assess extrapolation accuracy. The predicted computation time was T_{pred} = 1.1349 sec, while the measured time was T_{meas} = 1.1889 sec. Figure 12 (right-side) illustrates both values relative to the regression line, demonstrating that the prediction error remained within 4.6%. These results confirm that the computation time of PostGIS's Dijkstra-based route search can be reliably estimated in advance from network size using Equation (1). These results also quantitatively explain the substantial speedup achieved when detailed route computation is restricted to the target area in the proposed Time-Efficient MAS. As a result, limiting route computation to the target area reduced computation time to approximately 1/165 of the original (1.1889 sec ÷ 0.0072 sec), demonstrating a substantial improvement in processing efficiency. This reduction factor can be derived solely from the node and edge counts (V and E) of the respective networks. These trends were consistent across all four datasets, including the large-scale keikyu_db, indicating the robustness of the proposed model to different spatial structures.

5.2.2 Spatial Distribution of Co-Exist

This section reports an overview of the analysis results for Co-Exist, representing groups of agents that shared the same spatial location at the same time. The effects of transportation service reduction and population decline on Co-Exist were evident: the discontinuation of shared taxis and the reduction of bus services led to the disappearance of clusters within residential areas. In particular, the contraction of the bus network was shown to weaken the very structure of everyday contact opportunities, exerting a significant impact on the spatial connectedness of the local community (Figure 13).

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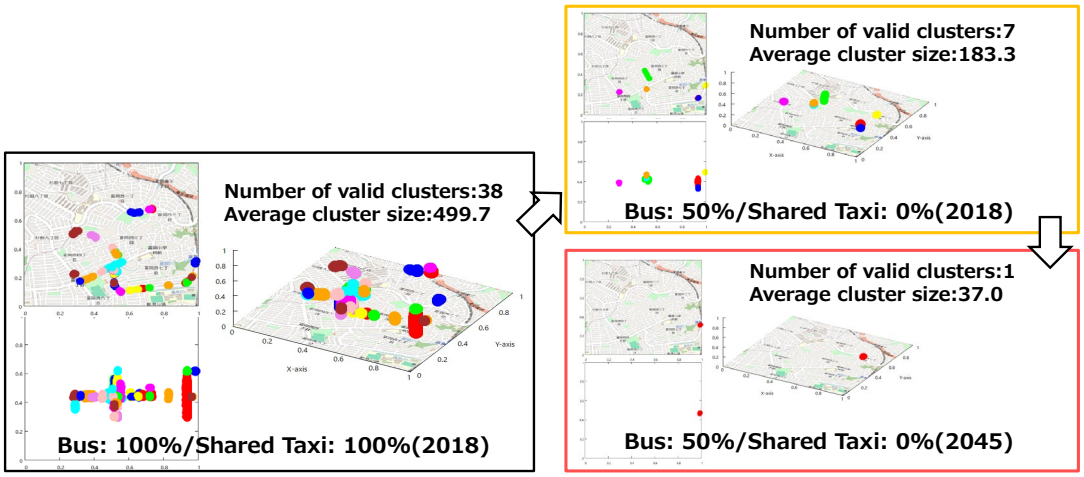


Figure 13: Changes in Co-Exist Clusters under Bus Service Reduction

5.2.3 Repeated Chance Meetings (RCM)

This section presents an overview of the analysis results for RCM, which represent spontaneous opportunities for contact occurring repeatedly between the same individuals.

(1) Comparison of RCM Counts Across Scenarios

This analysis compares the total RCM counts across all combinations of transportation policies and population scaling scenarios. First, regarding the effect of population scaling, all transportation scenarios under the 2045 projection show a decrease in RCM counts, which can be interpreted as a consequence of population decline leading to fewer overall opportunities for repeated opportunities for contact. Next, examining the impact of transportation policies, the overall trend indicates that the reduction of transportation services contributes to a decrease in RCM counts. However, under the 2018 population scale, the scenario combining bus service reduction and the abolition of shared taxis exhibited the highest RCM count. This result suggests that agents who lost access to public transportation shifted to walking-based mobility, thereby increasing the spatial dispersion of opportunities for contact and consequently amplifying the total number of RCM occurrences (Figure 14).

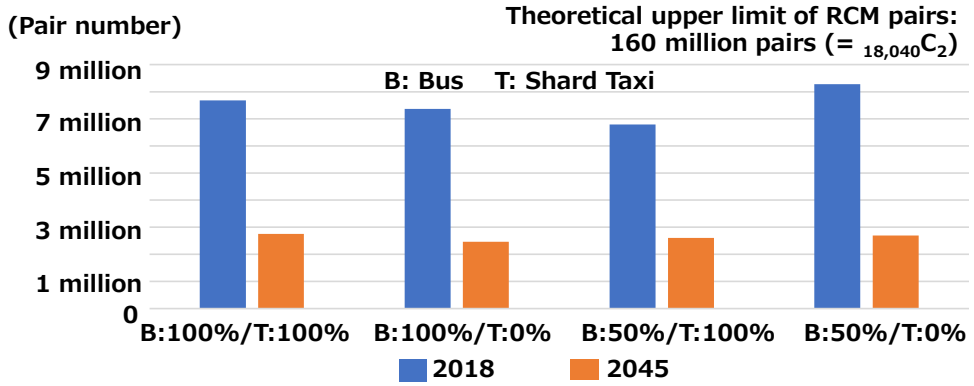


Figure 14: Comparison of Total RCM Counts Across Population and Transportation Scenarios

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(2) RCM Structure in the Baseline Scenarios for 2018 and 2045

This analysis presents the RCM Day, the number of days on which repeated opportunities for contact occurred ranked in descending order for agent pairs with the highest frequencies of RCM events. The results are shown separately for bus, shared taxi, and walking modes under the 2018 and 2045 baseline scenarios. In 2045, the number of RCM Day is lower than in 2018, which can be attributed to the effects of population decline. Moreover, in all transportation modes, the distribution tends to be left-skewed, indicating a concentration of opportunities for contact among a smaller number of agent pairs. This tendency is particularly pronounced in the shared taxi mode, where the analysis reveals a structure characterized by repeated interactions among a limited group of residents (Figure 15).

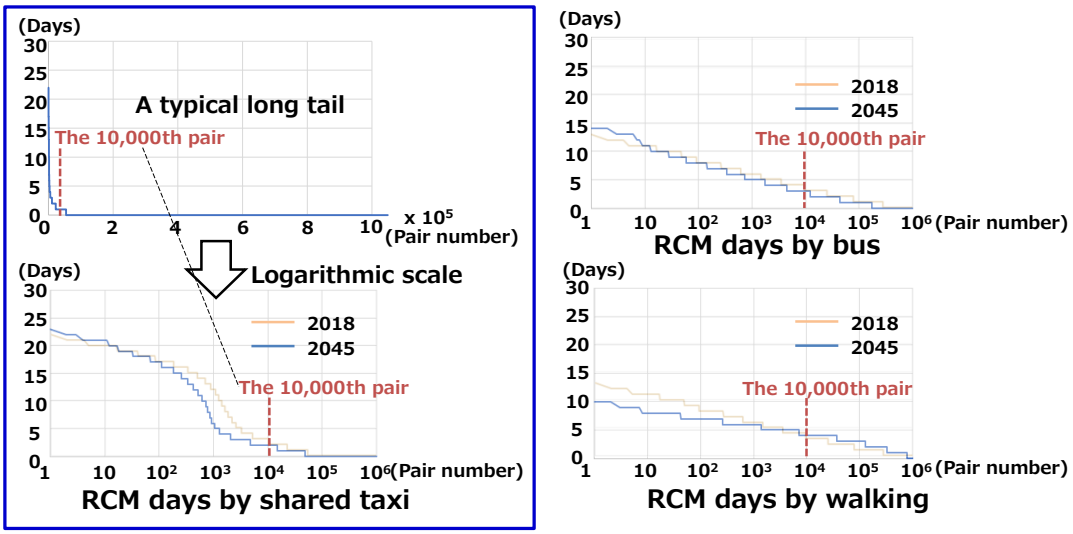


Figure 15: Distribution of RCM Day in the 2018 and 2045 Baseline Scenarios

(3) Structural Changes and Scenario Comparison of Same-Attribute Ratios

This analysis examines how the matching ratios of shared attributes specifically same district and same generation appear among the top-ranking pairs (those with the highest RCM counts). The comparison across scenarios reveals how different transportation policies and population scales influence the composition of repeated opportunities for contact in terms of spatial proximity and demographic similarity among residents (Figure 16).

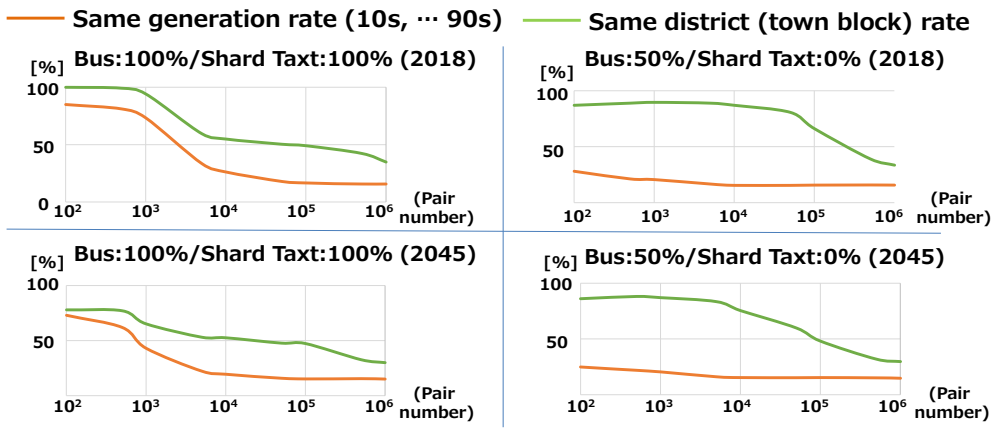


Figure 16: Comparison of Same-Attribute Ratios among Top RCM Pairs across Scenarios

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The results reveal two key findings:
(i) As expected, population decline leads to a reduction in overall contact opportunities; however,
(ii) An unexpected outcome emerges when bus and shared taxi services are discontinued, there is a tendency for contact opportunities within residential districts to increase.
This phenomenon can be interpreted as a side effect of the increase in walking-based mobility, where more residents move on foot within their neighborhoods, resulting in a higher likelihood of local opportunities for contact.

5.3 Interpretation of Simulation Results

This section interprets the simulation results presented in the previous sections, focusing on structural changes in Co-Exist and RCM across different scenarios. First, the analysis of Co-Exist distributions confirms that maintaining public transportation serves as a fundamental basis for sustaining residents' spatial opportunities for social interaction. Places where movement and stay intersect such as railway stations, bus stops, and commercial areas function not merely as transportation nodes but as potential arenas of social contact within the community. When these spaces are reduced, the total volume of contact declines, leading to a loss of diversity in spontaneous opportunities for contact. Next, the results of the RCM analysis indicate that the reduction of transportation services leads to qualitative changes in the structure of repeated opportunities for contact. As mobility options become limited, residents' activity ranges contract, and RCM occurrences become increasingly concentrated within the local community. While this may potentially strengthen cohesion within neighborhood communities, it simultaneously increases the risk of social isolation due to the loss of interactions with people outside the area. These results illustrate how changes in transportation services alter the spatial and temporal structure of Co-Exist and RCM, providing a basis for discussing their broader social implications.

6 Discussion and Policy Implications

6.1 Summary of Findings

Building on the interpretations of the simulation results presented in Section 5.3, this chapter discusses the broader implications of the findings for transportation policy and community structure. In this study, a Time-Efficient MAS was developed to quantitatively assess the impact of regional transportation policies on residents' social contact opportunities. The proposed method resolves the trade-off between computational cost and spatial resolution inherent in conventional MAS approaches, enabling large-scale behavioral reproduction within practical computation time. Through a case study of the Tomioka district, the following results were obtained.
(1) Demonstration of Time Efficiency: By introducing the route separation method, the proposed approach reduced computation time to 1/165 of that required by conventional methods, while enabling accurate estimation of total simulation time.
Based on (1), the following validations were conducted:
(2) Demonstration of Social Opportunity Restructuring: The spatiotemporal analysis of Co-Exist and RCM revealed that transportation policies significantly reshape the spatial distribution and recurrence structure of contact opportunities, with public transportation and walk-based mobility sustaining everyday encounters across time and location.
These findings present a new framework that extends the evaluation of transportation from "mobility efficiency" to "generation of social opportunities for contact," providing a foundation for visualizing the social value of transportation infrastructure.

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6.2 Implications for Urban and Mobility Policy

The results of the Time-Efficient MAS highlight the necessity of redefining transportation as a form of social infrastructure that mediates human relationships in regional communities. In particular, population decline and the reduction of transportation services were shown to decrease incidental opportunities for contact and weaken interactions beyond local neighborhoods, while simultaneously reinforcing localized community ties. These findings indicate that transportation policies influence not only mobility efficiency but also the structural configuration of community interactions. Accordingly, transportation policy should be discussed from a broader perspective that extends beyond conventional economic efficiency. In addition to criteria such as travel time, cost, and ridership, the ability of transportation systems to sustain opportunities for social contact should be explicitly considered. From this viewpoint, the following three aspects deserve priority in future urban and mobility policy discussions.

(1) Transportation Planning Incorporating Social Contact: Introduce a perspective that values not only travel efficiency but also the maintenance and promotion of Co-Exist and RCM opportunities.

(2) Emphasis on Community-Based Mobility: Position small-scale modes of mobility such as walking, buses, and shared taxis as "spaces of opportunity for contact" that sustain local interaction.

(3) Policy Evaluation through Digital Twins: Utilize the proposed framework to quantitatively assess how transportation policies influence SC and community connectivity.

This demonstrates a policy shift toward recognizing transportation as a social network infrastructure.

6.3 Limitations and Future Challenges

This study focuses on visualizing physical opportunities for contact and does not directly address psychological factors between agents, such as trust or preference. The calculated Co-Exist and RCM serve as quantitative indicators of potential opportunities for contact and do not guarantee the formation of social relationships themselves. From a computational perspective, although efficiency in route calculation was achieved, there remains room for improvement in memory optimization and clustering accuracy as the simulation scales up. Furthermore, the static behavioral data derived from the local PT survey have inherent limitations in capturing behavioral changes and responses to unexpected events. Future work should integrate dynamic data sources such as Mobile Spatial Statistics and Agoop datasets to enhance real-time responsiveness and behavioral diversity. Although the numerical reduction was confirmed in the Tomioka district case, the same computational characteristics were consistently observed across the four heterogeneous datasets used for parameter estimation. Because the proposed model depends only on the number of nodes and edges, similar efficiency gains are expected in other urban districts or policy scenarios with comparable spatial structures.

6.4 Directions for Future Research

Future research directions of this study can be summarized as follows.

(1) Introduction of Activity Agents: Extend the current agent model by probabilistically regenerating activity patterns to represent dynamic behavioral changes over time.

(2) Generalization and Comparative Analysis of Social Indicators: Establish Co-Exist and RCM as standardized social indicators by examining their relationships with SC and SWB, and by applying them to inter-city comparative studies.

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(3) Policy-Oriented Expansion with Dynamic Data: Develop the proposed framework into a practical policy support tool by integrating real-time mobility data, such as GPS-based human movement statistics, to enable scenario comparison and event-responsive evaluation.

7 Conclusion

This study developed a Time-Efficient Multi-Agent Simulation (Time-Efficient MAS) to quantitatively evaluate the social value of transportation policies in small urban areas while reflecting actual population structures. By resolving the trade-off between computational cost and spatial resolution inherent in conventional Multi-Agent Simulations (MAS), the proposed framework enables long-term, population-scale simulations to be executed within practical computation time. Resident agents were generated based on Person Trip (PT) survey data, incorporating real population attributes such as place of residence, age, gender, and trip purpose. Individual travel routes were reconstructed using GIS data, and a route calculation method that separates the target area from the surrounding area was introduced to significantly reduce computational load. As a result, large-scale and high-resolution simulations of resident mobility and behavior were achieved.

To analyze the social dimension of mobility, this study introduced two novel indicators: Co-Exist, representing spatiotemporal coexistence arising from movement and stay, and RCM, capturing the recurrence of incidental opportunities for contact between the same individuals. These indicators enable visualization of the structure of opportunities for contact formed through everyday mobility and allow quantitative assessment of how transportation policies influence the frequency and spatial distribution of social interactions.

A case study of the Tomioka district demonstrated that reductions in public transportation services and the progression of population aging lead to increased reliance on walking-based mobility. This shift localizes residents' activity ranges and decreases the frequency of RCM, indicating that transportation policies affect not only travel efficiency but also the continuity of social connections and the structure of local communities.

Future research will advance the proposed framework in three directions. First, activity-based agents will be introduced to represent dynamic behavioral adaptations, including changes in preferences and decision-making. Second, Co-Exist and RCM will be examined in relation to SC and SWB to establish a multidimensional framework for evaluating the social value of mobility. Third, the applicability of the proposed method will be extended through inter-city comparative studies, clarifying how regional characteristics and demographic composition shape patterns of social interaction.

This study addressed two central objectives: the establishment of Co-Exist and RCM as quantitative indicators of social interaction, and the development of a Time-Efficient MAS capable of executing month-scale simulations involving approximately 18,000 agents. The results confirmed that the proposed computational framework enables practical large-scale simulations, and that its regression-based performance model allows route computation time to be predicted in advance from network size. At the same time, the Co-Exist and RCM indicators provide a foundation for analyzing how mobility policies shape the structure of social interactions within communities.

Beyond transportation, the proposed framework is applicable to the analysis of service systems in which social interaction emerges through use. Co-Exist and RCM function as knowledge representations of interaction opportunities generated by services, supporting service design and management that consider the continuity of human interactions as a dimension of social value.

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The Time-Efficient MAS proposed in this study provides an analytical framework for redefining transportation not merely as a mobility infrastructure but as a social foundation that shapes opportunities for human contact. By enabling quantitative exploration of how people meet and interact through everyday mobility, this framework contributes knowledge for policy design and decision-making aimed at sustaining socially connected urban communities.

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